P-S-651E
May 14, 1974
SUPERSEDING
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May 21, 1968

FEDERAL SPECIFICATION

SODIUM ORTHOSILICATE, TECHNICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one technical grade of sodium orthosilicate which is an intimate mixture of caustic soda and sodium metasilicate.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin.
PPP-D-723 - Drums, Fiber.
PPP-P-45 - Packaging of Bulk Quantities of Soaps, Detergents and Related Products (Chip, Powdered, or Granular Form) For Domestic and Overseas Shipment.
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 Through 12 Gallon).

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 536 - Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing.

FSC 6810

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-15011 - Pallets, Material Handling, Wood, Post Construction, 4-Way Entry.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

Code of Federal Regulations.

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

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(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171-179.")

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

National Classification Board:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

American Society for Testing and Materials (ASTM) Standards:

D501-67 - Sampling and Chemical Analysis of Alkaline Detergents.
D1193-70 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

American Pharmaceutical Association Publication:

The National Formulary - Thirteenth Edition.

(Application for copies should be addressed to the Mack Publishing Co., Easton, Pennsylvania 18042.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Material. Sodium orthosilicate shall be an intimate mixture of caustic soda and sodium metasilicate in the form of free-flowing white granules or flakes which have been treated with an anti-dusting oil so as to minimize air-borne particles during handling. The anti-dusting oil shall be a light mineral oil conforming to the National Formulary requirements for Light Mineral Oil except that is shall have a specific gravity of 0.820 to 0.850 at 77° F and a viscosity of 50 to 100 Saybolt Universal seconds at 100° F.

3.2 Chemical characteristics. Sodium orthosilicate shall conform to the chemical characteristics of table I when tested as specified therein.

Table I - Chemical characteristics

Characteristic	Percent by weight: Minimum	Percent by weight: Maximum	Test paragraph
Total alkalinity (as Na_2O)	60	----	4.2.4.1
Total silica (as SiO_2)	27	----	4.2.4.1
Water-insoluble matter	----	0.6	4.2.4.1
Sodium carbonate (Na_2CO_3)	----	4.5	4.2.4.2
Anti-dusting oil	0.20	0.30	4.2.4.3

3.3 Particle size. No less than 99 percent by weight of the sodium orthosilicate shall pass through a 2.38-millimeter (No. 8) sieve and no less than 95 percent by weight shall be retained on a 177-micron (No. 80) sieve when tested as specified in 4.2.4.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

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4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the sodium orthosilicate produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 1 pound (lb) shall be removed from each sample container and placed in a suitable closed, clean, dry container labeled to identify the lot and container from which it was taken (see 6.5).

Table II - Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled shipping container ready for shipment. Sample shipping containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Bag liner missing or not as specified (where required)
(f) Marking incorrect, missing, or illegible
(g) Palletization not as specified

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4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Total alkalinity (as Na_2O), total silica (as SiO_2), and water-insoluble matter. Determine percent total alkalinity, percent total silica, and percent water-insoluble matter in the specimen in accordance with ASTM D501 under the test methods for Sodium Metasilicate, Sodium Sesquisilicate, and Sodium Orthosilicate.

4.2.4.2 Sodium carbonate. Determine percent sodium carbonate in accordance with Fed. Test Method Std. No. 536, method 1401 except that 50 milliliters (ml) of freshly boiled water and 60 ml of 1 to 1 sulfuric acid shall be substituted for the water and hydrochloric acid specified in method 1401.

4.2.4.3 Anti-dusting oil. Weigh an extraction flask. Carefully weigh 100.0 $\pm$ 0.1 gram of specimen into a tared Soxhlet extraction unit paper thimble 120 millimeters (mm) long by 40 mm in diameter and extract with chloroform in a Soxhlet extraction unit (2 cycles) for at least 3 hours. Remove the flask containing all the extraction solvent and evaporate the extraction under a hood. Dry the residue to constant weight in an oven at 105° $\pm$ 2° C. (Note: Care must be used in pouring the solvent into the extraction tube and in removing the paper thimble from the tube after extraction so as not to allow any of the specimen particles to escape from the thimble into the solvent.) Calculate the percent by weight anti-dusting oil as follows:

$$\text{Percent anti-dusting oil} = \frac{100A}{W}$$

where: A = Weight of residue in grams and
W = Weight of specimen in grams.

4.2.4.4 Particle size. Determine particle size in accordance with Fed. Test Method Std. No. 536, method 2101. If a constant temperature and humidity room is not available and if the relative humidity is greater than 50 percent, the particle size test shall be completed as quickly as possible.

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5. PREPARATION FOR DELIVERY

5.1 Packing. Sodium orthosilicate shall be packed level A, B, or C as specified (see 6.2).

5.1.1 Level A. Sodium orthosilicate shall be packed in 25-pound or 100-pound quantities as specified (see 6.2).

5.1.1.1 Twenty-five-lb quantity. A quantity of 25 (+1/4 or -0) lb of sodium orthosilicate shall be packed in 3 gallon (gal) nominal capacity metal pail conforming to type II, class 3 of PPP-P-704 in accordance with any applicable Department of Transportation (DOT) regulations. Each pail shall be provided with a close-fitting bag liner formed from polyethylene film of 0.002 inch minimum thickness conforming to type I, class 1, grade and finish optional of L-P-378. All seams, except that of the closure, shall be heat sealed and shall meet the heat-seal seam requirements of L-P-378. The bag liner shall be closed by heat sealing as specified in L-P-378 or by tying or knotting.

5.1.1.2 One hundred-lb quantity. A quantity of 100 (+1 or -0) lb of sodium orthosilicate shall be packed as specified for the 25-lb quantity in 5.1.1.1 except that the pail shall have a nominal capacity of 10 gal and shall conform to type II, class 5 of PPP-P-704 and the thickness of the material used for the bag liner shall be no less than 0.004 inch.

5.1.2 Level B. Sodium orthosilicate shall be packed in 25-pound or 100-pound quantities as specified (see 6.2).

5.1.2.1 Twenty-five-lb quantity. A quantity of 25 (+1/4 or -0) lb of sodium orthosilicate shall be packed as specified in 5.1.1.1 except that a fiber drum shall be used instead of the metal pail and in addition to the bag liner, each drum shall be furnished with a 0.001 inch minimum thickness polyethylene film moisture barrier incorporated in one of the inner plies of the side wall. The fiber drum shall have a nominal capacity of 3 gal and shall conform to DOT Specification 21C and to type III, grade A, class 1 of PPP-D-723.

5.1.2.2 One hundred-lb quantity.

(a) For military activities. A quantity of 100 (+1 or -0) lb of sodium orthosilicate shall be packed as specified for the 25-lb quantity in 5.1.2.1 except that the nominal capacity of the drum shall be 10 gal and the thickness of the material used for the bag liner shall be no less than 0.004 inch.

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(b) For civil agencies. A quantity of 100 lb of sodium orthosilicate shall be packed in a fiber drum conforming to PPP-P-45 for level B packing requirements.

5.1.3 Level C. A quantity of 25 (+1/4 or -0) lb or 100 (+1 or -0) lb as specified (see 6.2) of sodium orthosilicate shall be packed in a container constructed to protect the sodium orthosilicate from alteration and to assure its safe delivery without damage. The container shall comply with any applicable DOT regulations and Uniform Freight Classification rules or National Motor Freight Classification rules, as applicable. The containers shall be sufficiently rigid to withstand transportation and to permit them to be stored in tiers at least 10 feet high without damage to containers or contents. The gross weight and size limitations of the containers shall not exceed the tariff or classification limitations specified and the containers shall be closed or strapped as specified therein.

5.2 Marking. Marking for civil agencies shall be in accordance with any applicable DOT regulations and Fed. Std. No. 123. Marking for military activities shall be in accordance with any applicable DOT regulations and MIL-STD-129. Each shipping container shall be durably and legibly labeled to show the following precautionary information:

WARNING!
CAUSES SEVERE BURNS TO SKIN AND EYES
HANDLE WITH CARE
KEEP CONTAINER CLOSED WHEN NOT IN USE

Avoid contact with skin, eyes, and clothing.
Do not take internally.
When handling, wear rubber gloves and goggles or face shield.
While making solutions, add slowly to the surface of the solution to avoid violent spattering.
Do not add to solutions over 120° F.

In case of contact:

with skin - immediately flush skin with plenty of water.
with eyes - immediately flush with plenty of water and continue for at least 15 minutes; get immediate medical attention.

5.3 Palletization. Unless otherwise specified in the contract or order (see 6.2), uniform quantities of shipping containers shall be palletized. For level A and level B protection, shipments shall be palletized in

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accordance with the applicable requirements for load type III of MIL-STD-147 for pails and in accordance with load type IV of MIL-STD-147 for fiber drums. The pallet shall conform to style 1, type I, class 1 of MIL-P-15011. For level A shipments, steel strapping shall be used and the pallet and wood storage aids, including the cap and tray, shall be treated as specified for the pallet in MIL-P-15011. For level B shipments, a fiberboard cap shall be used and wood parts used in palletization need not be treated as specified for pallets used for level A shipments. For level C shipments, any palletization shall be permitted provided that it conforms to Uniform Freight Classification rules or National Motor Freight Classification rules and that it permits stacking of palletized shipping containers in tiers at least 10 feet high without damage.

6. NOTES

6.1 Intended use. Sodium orthosilicate is intended for laundry, metal cleaning (except aluminum), and heavy duty cleaning purposes.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Unit quantity required (see 5.1.1, 5.1.2, and 5.1.3).
(c) Level of packing required (see 5.1).
(d) If palletization is not required (see 5.3).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

6.5 Sampling precaution. Because of the hygroscopic nature of sodium orthosilicate, care should be taken to keep exposure of the samples to air to a minimum.

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CIVIL AGENCIES COORDINATING ACTIVITIES:

COM - NBS
GSA - FSS
LAB
VA - DMS

MILITARY INTEREST:

Custodians:

Army - EA
Air Force - 68

Review activities:

Army - GL, MD, ME
DSA - GS

User activities:

Army - SM
Navy - AS

Preparing activity:

Army - EA

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

P-S-651E - Sodium Orthosilicate, Technical

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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